ABDEL-RAHMAN NOBI

Senior Mechanical Design & R&D Engineer

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Open to International Relocation | Europe & GCC

PROFESSIONAL SUMMARY

Senior Mechanical Design and R&D Engineer with 6+ years in product design and R&D and 7 years in engineering. I take mechanical products from first sketch to series production: 3D design in Autodesk Inventor and SolidWorks, GD&T and DFM/DFA, simulation in ANSYS, physical drop and IP testing, and iLogic design automation. I designed a medium-power busway product line from concept to series production, and busway for 30+ projects across Egypt, Saudi Arabia and the UAE, including 6 km+ for the sewage treatment plant at King Salman Park. I design to IEC 61439, UL 857 and the NEC, and I review and sign off the work of junior engineers before it reaches the factory.

CORE COMPETENCIES

Mechanical Product Design • Product Development (Concept to Series Production) • R&D & Design Validation • Sheet Metal & Plastic Part Design • GD&T & Tolerancing • DFM / DFA • FEA & Explicit Dynamics Simulation • Drop & IP Testing • Failure & Root-Cause Analysis • iLogic Design Automation • BIM Coordination (Revit / BIM 360) • BOM & Engineering Release • Engineering Leadership & Mentoring

PROFESSIONAL EXPERIENCE

Senior Design Engineer | Elsewedy Electric — Jun 2025 – Present

Cairo, Egypt

- Designed a new medium-power busway product line from first concept to series production: aluminium housing and copper or aluminium conductors engineered to each client's specification, simulated in ANSYS and designed to IEC 61439-1, IEC 61439-6 and UL 857.
- Made the new product line more cost-competitive by localising its materials in Egypt.
- Built an iLogic configuration framework in Autodesk Inventor: one master model generates rule-checked parts, assemblies and drawings for every product variant, replacing repetitive manual modelling.
- Lead mechanical design of busway and busbar systems for industrial, commercial and data-centre projects; review and sign off junior engineers' drawings and BOMs before release, and mentor the team.
- Work with manufacturing, procurement and QA from design through production, and support installation and commissioning on site, resolving field issues through root-cause analysis.

Design Engineer | Elsewedy Electric — May 2024 – May 2025

Cairo, Egypt · Promoted to Senior Design Engineer after 13 months

- Designed 6 km+ of busway for the sewage treatment plant at King Salman Park, Riyadh: layouts in Revit, coordination with every discipline through BIM 360, and manufacturing item lists for each run.
- Designed 1.5 km+ of busway for the Warner Bros. World Abu Dhabi Phase 2 expansion, from layouts to detailed drawings and BOMs.
- Delivered busway design and BIM coordination for Khazna Data Centers in the UAE.
- Turned customer specifications into layouts, detailed drawings and BOMs released through Oracle ERP / PLM; prepared hanger support verification calculations and supply offers from client BOQs.

Mechanical Design Engineer | FemtoPay — Oct 2022 – Mar 2024

Cairo, Egypt

- Designed enclosures for new electronic products in plastic and metal, taking them from concept to mass production.
- Defined injection-mould tooling and manufacturing requirements, and detailed assemblies with supporting calculations.
- Ran drop-impact and IP ingress tests myself and redesigned parts until every design passed.
- Optimised designs for function, reliability and manufacturability with manufacturing, quality and project teams, and proposed cost reductions.

Mechanical R&D Officer | Egyptian Armed Forces — Jun 2020 – Sep 2022

Cairo, Egypt

- Led the R&D team on mechanical and electrical systems for air-defence applications: reliability testing, structural analysis and performance optimisation.
- Wrote failure-analysis reports with root causes and corrective actions to keep equipment operational.

Mechanical Site Engineer | IETOS — Aug 2019 – May 2020

Luxor, Egypt

- Built and installed the mechanical systems of a sewage treatment plant (activated sludge process): project schedules, subcontractor coordination and variations.

TECHNICAL SKILLS

CAD & BIM: Autodesk Inventor (primary, iLogic), SolidWorks, CATIA V5, AutoCAD, Revit / BIM 360

Simulation & Analysis: ANSYS Mechanical, ANSYS Explicit Dynamics, LS-DYNA, hand calculations

Product Engineering: DFM / DFA, DFMEA, GD&T, sheet metal, plastics and injection moulding, prototyping, drop and IP testing, root-cause analysis

Automation & Systems: iLogic, Excel, VBA, Python, Oracle ERP / PLM, KeyShot

STANDARDS & CODES

IEC 61439-1 • IEC 61439-6 • UL 857 • NFPA 70 (NEC) Articles 368 & 370 • IEC 60529 • GD&T

EDUCATION & CERTIFICATION

B.Sc. Mechanical Engineering — South Valley University, Egypt | 2014 – 2019

ASME GDTP Senior (Y14.5-2018) — in progress

PUBLICATIONS & PORTFOLIO

“Diving into Chaos: Explicit Dynamics” — LinkedIn article, April 2024: ANSYS Explicit Dynamics drop-test simulation and solver selection.

Portfolio: abdelrahmannobi.com — projects, product designs and case studies.

LANGUAGES & ACTIVITIES

Arabic (native) • English (fluent) • German (beginner)

IEEE SAC Egypt member (2014 – 2017) • Microsoft Student Partner (2015 – 2017)